

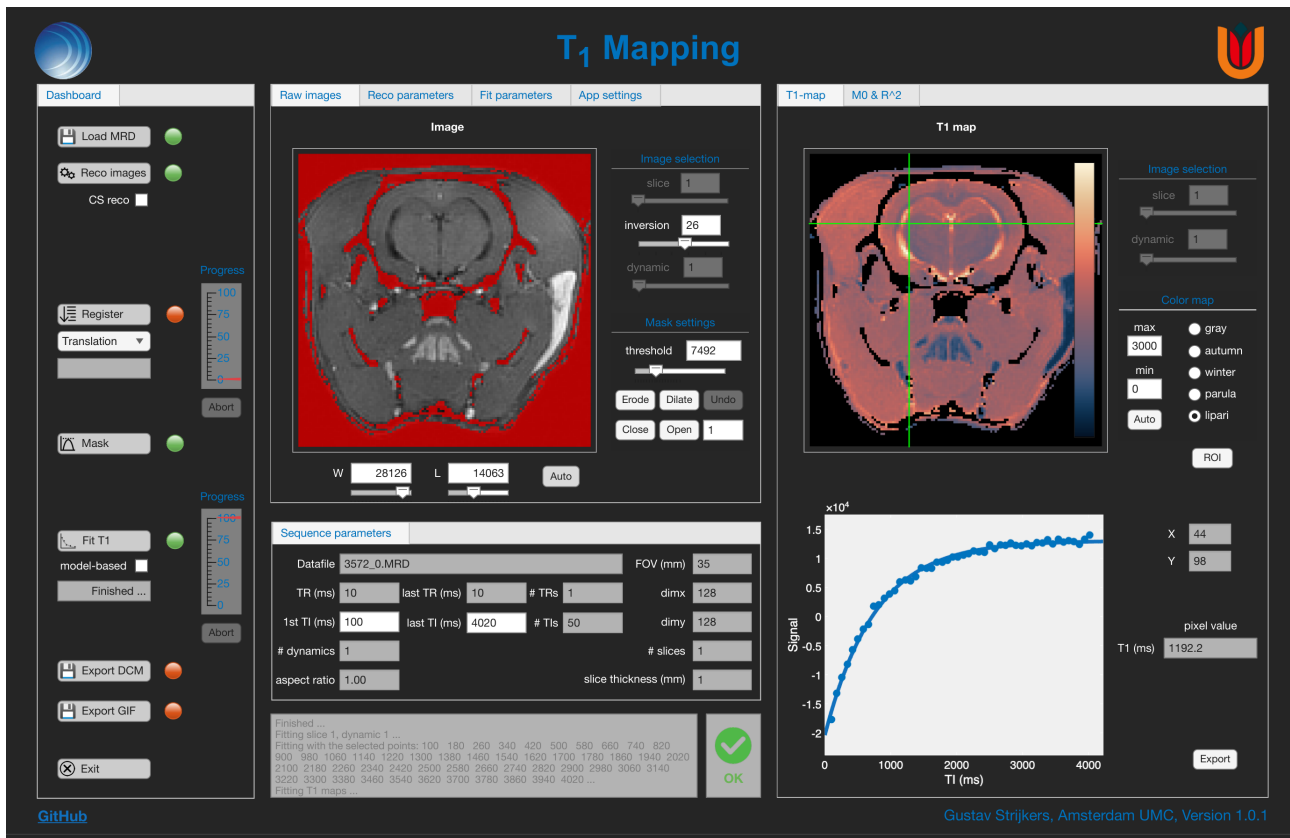


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1. Overview

T1mapp is a software utility and graphical user interface to reconstruct T1 maps from inversion recovery (Look-Locker), saturation recovery or variable flip angle experiments acquired with an MR Solutions preclinical MRI system.

2. Contact

T1mapp and this manual are written by:

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3. Disclaimer

The software has been extensively tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system.

However, this does not warrant the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free.

In case of questions or issues, please contact Gustav Strijkers.



4. Installation notes

Software download

The Matlab app and a Windows standalone version (using the free Matlab runtime engine) can be downloaded from GitHub:

<https://github.com/Moby1971?tab=repositories>

Installation of the Windows standalone version

`T1mapp_app_installer.exe`

Will install the Matlab runtime engine and the *T1mapp* program.

Bart toolbox download

Compressed sensing reconstruction and model-based fitting in T1mapp require the Bart reconstruction toolbox, version 1.0.00 or newer, which can be downloaded from:

<https://mrirecon.codeberg.page/>

Without Bart, or with an older version, both are switched off, and the message box says so at start. When Bart is built with GPU support, T1mapp runs them on the GPU.

Elastix registration

Image registration uses elastix, which comes with T1mapp for macOS, Windows and Linux. The macOS version is built for Apple silicon; on an Intel Mac, which is no longer supported, it does not run. When elastix cannot be run, T1mapp registers the images with Matlab's `imregister` instead, and says so at start.



Bart toolbox installation in OSX with MacPorts

(1) Install Xcode from the Mac App Store

```
$ xcode-select --install
```

(2) Install MacPorts (<http://www.macports.org/>)

It is recommended to install a newer version of gcc from MacPorts

(3) Install packages

```
$ sudo port install fftw-3-single
$ sudo port install gcc14
$ sudo port install libpng
$ sudo port install openblas
$ sudo port install flock
$ sudo port install gmake
$ sudo port install coreutils
$ sudo port install libblocksruntime
```

(4) Make

```
$ export PATH=/opt/local/libexec/gnubin:$PATH
$ CC=gcc-mp-14 gmake
$ sudo gmake install
```

Bart toolbox installation in OSX with HomeBrew

(1) Install Xcode from the Mac App Store

```
$ xcode-select --install
```

(2) Install HomeBrew

```
$ /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

(3) Install packages

```
$ brew install libpng
$ brew install fftw
$ brew install openblas
$ brew install gmake
$ brew install llvm libomp
```

(4) Make

```
$ CC=gcc MACPORTS=0 gmake
$ sudo gmake install
```



Bart toolbox installation in Windows 11

(1) Install the Windows subsystem for Linux

Start a windows powershell as administrator and run the following command:

```
Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Windows-Subsystem-Linux
```

A system restart will be needed.

For more information, see:

<https://pureinfotech.com/install-wsl-windows-11/>

(2) Download Ubuntu Linux

Open PowerShell

To see a list of available Linux distributions for download, enter:

```
wsl -l -o
```

Install a Linux distribution:

```
wsl --install -d <distribution name>
```

You will be asked to create a user account the first time you start a Linux command prompt.

(3) Upgrade to WSL2

(see e.g. <https://pureinfotech.com/install-wsl-windows-11/> on how to upgrade).

(4) Upgrade the Linux distribution

```
$ sudo apt-get update
$ sudo apt-get dist-upgrade
```

(5) Install Bart prerequisites (gcc 12 or newer)

```
$ sudo apt-get install make gcc libfftw3-dev liblapacke-dev
libpng-dev libopenblas-dev gfortran
```

(6) Download Bart

```
$ wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
```

(7) Build Bart

```
$ tar xvfz v1.0.00.tar.gz
$ cd bart-1.0.00
$ make
$ sudo make install
```



5. Running the software

Running in Matlab 2025a

The T1mapp software can be started from its root directory from the command line.

```
>> T1mapp
```

T1mapp starts in about 10 seconds. A 3D variable flip angle scan of 128 x 96 x 96 with 12 flip angles needs about 3 GB of memory, an inversion recovery scan of 128 x 128 with 50 inversion times and 25 dynamics about 3.5 GB.

Additional licenses may be required.

```
MATLAB  
Image Processing Toolbox  
Signal Processing Toolbox
```

Running the Windows standalone

The Windows standalone version can be run from the start menu or the desktop icon.

Settings and log files

T1mapp keeps its settings in a file, t1mappSettings.json, in a folder of its own for every version of the app:

```
macOS      ~/Library/Application Support/T1mapp/<version>  
Windows    %APPDATA%\T1mapp\<version>
```

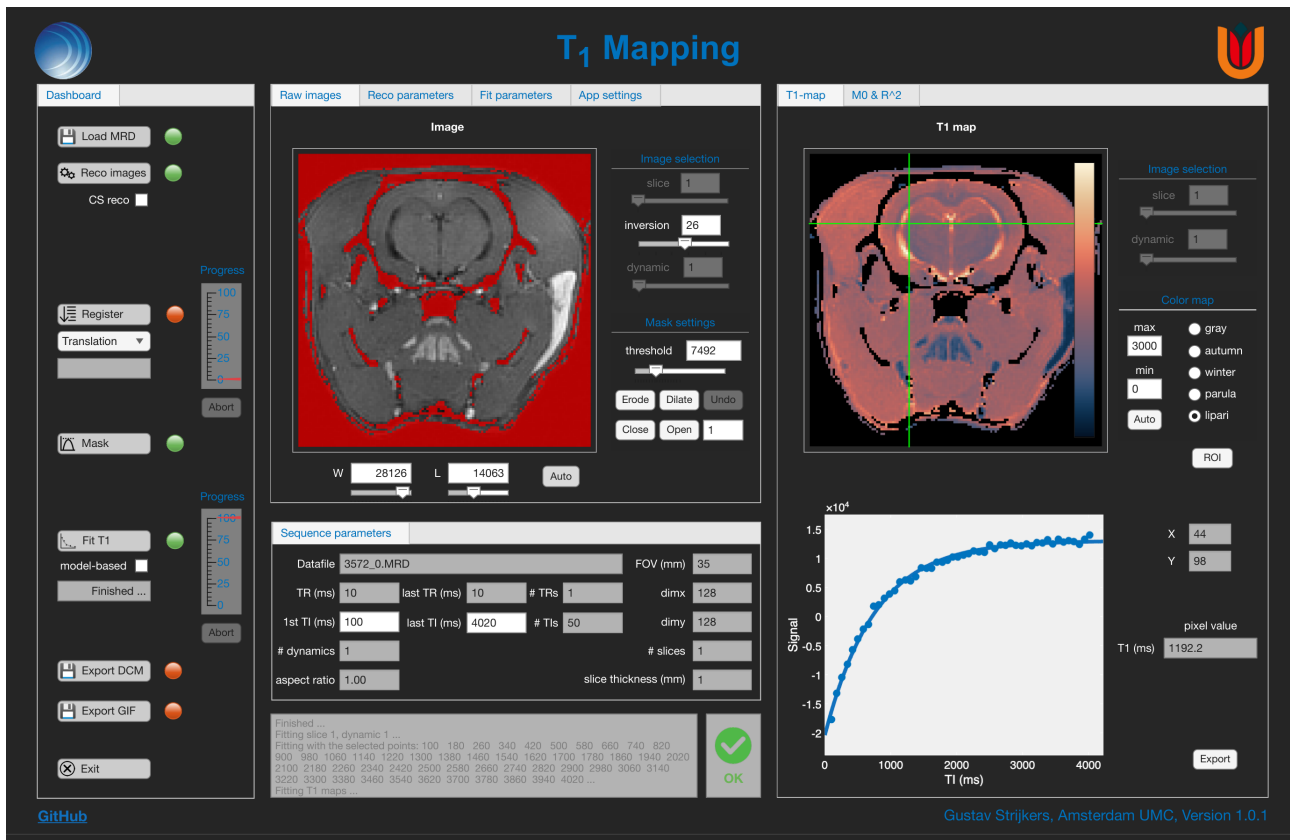
The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds whether the app looks for a newer version, the folders where loading and the exports start, the color scheme, the sounds, the reordering of segmented k-space, and the values the reconstruction, registration, fit and display fields start with. A value that cannot be used is replaced by its standard value, and the message box says so. The color scheme and the sounds can also be switched in the App settings tab, for the session only.

Every session writes a log file, t1mapp-log-<date and time>.txt, to the same folder. The log keeps every message and the parameters of each import, reconstruction, registration and fit. Logs older than the newest 50 are removed at start.



6. Basic operation

The T1mapp program operates from a single window with 5 panels.



Panel 1: Dashboard

This panel contains the task buttons and parameters that control the reconstruction process. The tasks run from top to bottom; registration and masking can be left out. A green light shows a task done, yellow a task running, and red one not done or failed.

Panel 2: Raw images - Reco parameters - Fit parameters - App settings

This panel has 4 tabs: raw images, reco parameters, fit parameters and app settings.

Panel 3: Sequence parameters

This panel displays the relevant acquisition parameters; for inversion recovery the first and last TI can be edited, which spaces the TIs again and clears a fit.

Panel 4: T1-map - M0 & R²

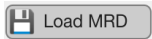
During and after fitting this panel shows the T1 maps. The M0 and R² goodness-of-fit maps are displayed in the second tab.

Panel 5: Messages

Displays program messages and the status: OK, BUSY, WARNING or ERROR.



Step 1: Loading data

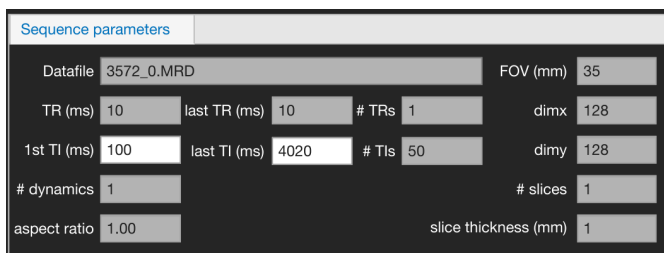
Press  to load the MRD raw data file.

T1mapp recognizes three T1 measurements from the MRD parameters: inversion recovery (Look-Locker FLASH), saturation recovery (multi-TR) and variable flip angle series; other scans are refused, with the reason in the message box.

If a segmented FLASH scan does not reconstruct properly, tick toggle reorder in the App settings tab and load it again.

For multi receiver coil data, select 1 of the MRD raw data files.

Relevant acquisition parameters will be shown in panel 3.



Sequence parameters

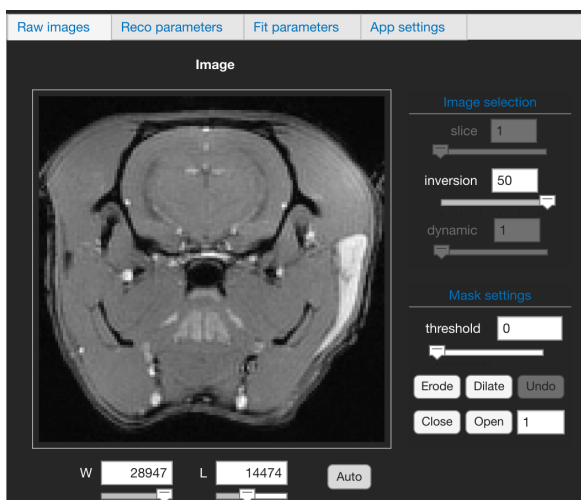
Datafile	3572_0.MRD			FOV (mm)	35
TR (ms)	10	last TR (ms)	10	# TRs	1
1st TI (ms)	100	last TI (ms)	4020	# TIs	50
# dynamics	1			# slices	1
aspect ratio	1.00			slice thickness (mm)	1
				dimx	128
				dimy	128

Step 2: Image reconstruction

Press  to reconstruct again, for instance with compressed sensing; loading reconstructs the images already.

The standard reconstruction is a 2D or 3D FFT with PCA denoising, on by default.

When the CS reco checkbox is checked and Bart is available, the images are reconstructed by compressed sensing with Bart, with the weights of the Reco parameters tab: wavelets (WV), total variation (TV) and locally low rank (LLR), in space (xyz) or along the series (TVti, TVtr or TVfr) and the dynamics (TVdyn or TVfa). That tab also switches PCA denoising and Gibbs ringing suppression (off by default).



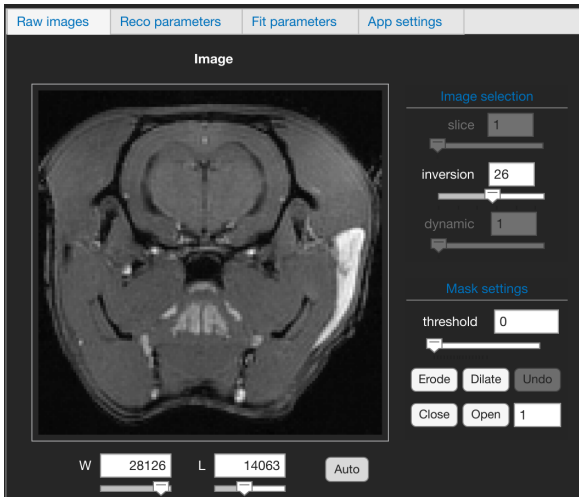
Images with different inversion and dynamic times can be inspected with the sliders.



Step 3: Image registration

NOTE: this step usually can be omitted.

Use the sliders and/or edit-fields to inspect the images as function of slice (Slice), inversion time (Inversion), or repetition (Dynamic). The window and level can be set with the W and L fields and sliders, or with Auto.



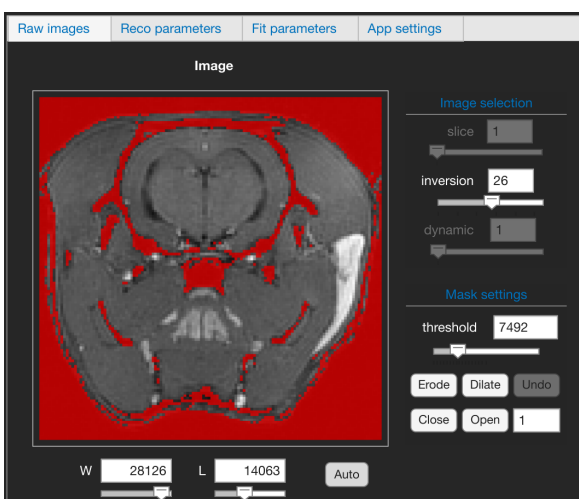
In case the images as function of the inversion times or dynamics are not perfectly registered, press  **Register** to register them to the last inversion time (the longest TR, or the brightest flip angle) with elastix, as a translation, rigid, affine or B-spline transform (the list below the button). Abort stops it. Registering clears the mask and the maps, and rules out the model-based fit.

Step 4: Masking the background

To prevent unnecessary fitting of the background pixels, the images need to be masked first; without a mask every pixel is fitted, the background included.

Press  **Mask** to mask them with an automatic (Otsu) threshold per slice.

The result will look something like the figure below in which the red pixels indicate the regions that will be skipped during T1 fitting.



In case the automatic threshold is not satisfying, it can be adjusted per slice with the threshold field or slider. The mask can be refined by erode, dilate, close and open operations, of the size in pixels in the field beside them; Undo takes the last one back.



Step 5: Fitting T1 maps

Press the Fit parameters tab to inspect the fit settings.

The screenshot shows the 'Fit parameters' tab with the following settings:

- Slice selection:** first slice: 1, last slice: 1
- Dynamic selection:** first dynamic: 1, last dynamic: 1
- Goodness of fit:** min R-square: 0.80
- Inversion time selection:** A grid of 96 checkboxes (8 rows by 12 columns) all of which are checked.

The first and last slice and dynamic that will be fitted can be selected.

Min R-square is the lowest R^2 accepted: where the fit of a pixel has a lower R^2 , or gives a T1 above 10 s, the T1 and M0 values are set to zero. This holds for the least-squares fits.

The bottom part of this tab lists the points of the series in the order of the scan (inversion times, TRs or flip angles): those ticked are fitted, so points can be left out, as long as 3 remain.

Press  to start the T1 fitting; Abort on the dashboard stops it.

For inversion recovery the fit is a least-squares fit of the equation

$$\text{Signal} = |A - B \exp(-TI/T1^*)|$$

$$T1 = (B/A - 1)T1^*$$

Model-based fitting

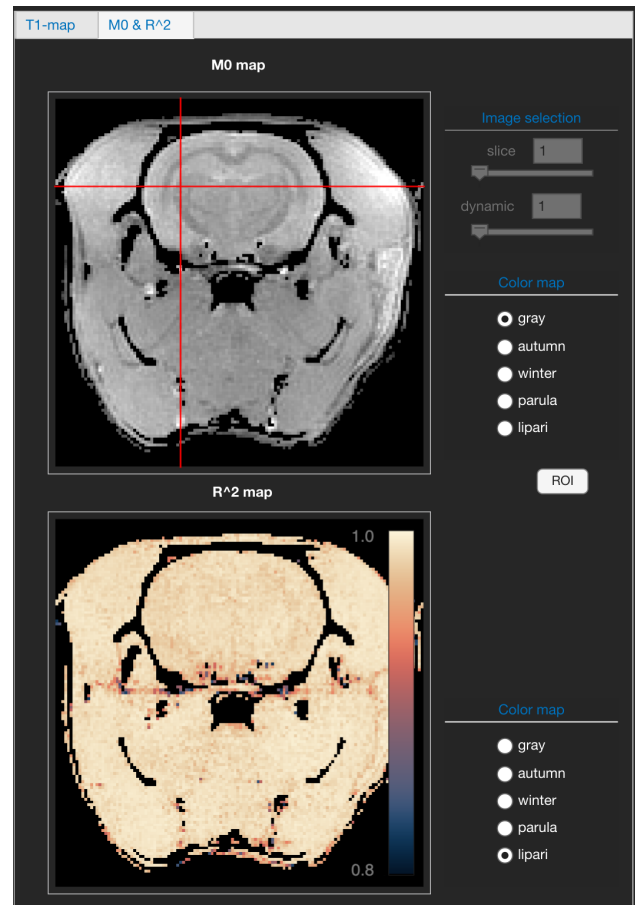
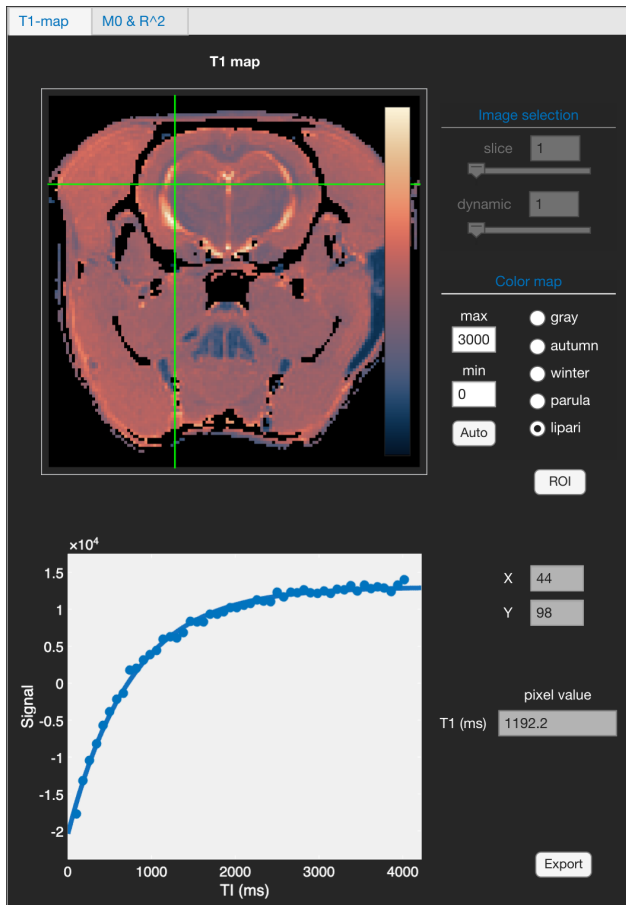
Tick the model-based checkbox  to fit a model-based reconstruction with Bart.

The T1 map is then fitted to the k-space directly. It is offered for 2D inversion recovery that has not been registered.



Step 6: Fit results

The T1 maps will be shown in the T1-map tab, on the scale set with max and min, or with Auto. An M0 map and an R² map (for least-squares fitting only) are shown in the second tab. A different color map can be chosen for each. After the fit, the T1 scale runs from 0 to twice the median T1, and the sliders beside the maps step through their slices and dynamics.

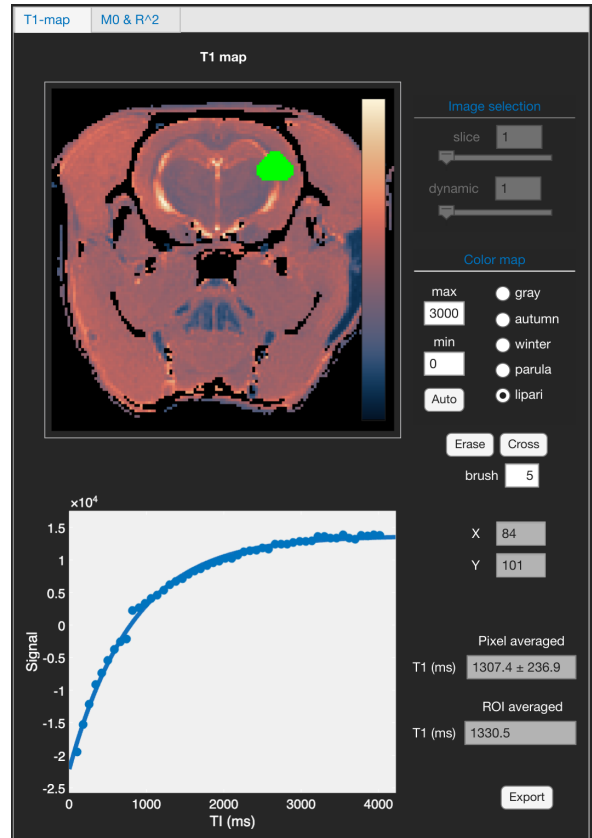
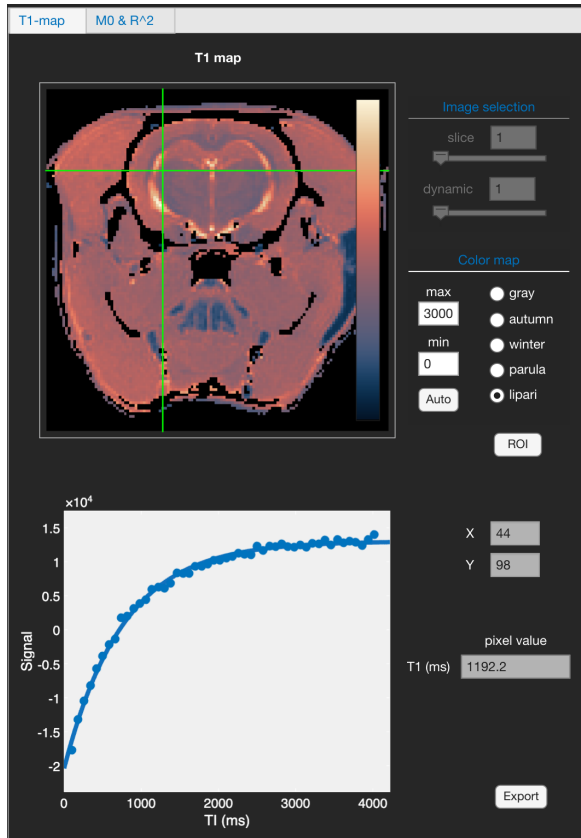




Step 7: Extracting T1 values from pixels and ROIs

When you click on the T1 map a cross-hair will be shown, with the signal of that pixel as a function of T1 (symbols), the fitted curve (line) and its T1 value. The signal of an inversion recovery is shown with the sign the fit restored.

A red symbol indicates that this T1 was omitted in the fitting process, according to the list in the Fit parameters tab (see step 5).



You can toggle between the cross-hair and region-of-interest (ROI) values by pressing the **ROI** button. A single ROI can be 'painted' on the T1 map in one or more slices by mouse clicks on the image, with the brush size in pixels set below the button.

For the ROI, two T1 values are calculated:

- Pixel averaged: The average of the T1 values of the pixels, with their spread.

$$T1_{ROI} = 1/\text{mean}(R1_{\text{pixel}})$$
- ROI averaged: The T1 equation is fitted to the average signal intensities in the ROI, producing a single T1 value for that ROI.

The ROI can be removed with **Erase** button.

The ROI-averaged signal is written to a CSV file with **Export** button, beside the inversion times, TRs or flip angles.

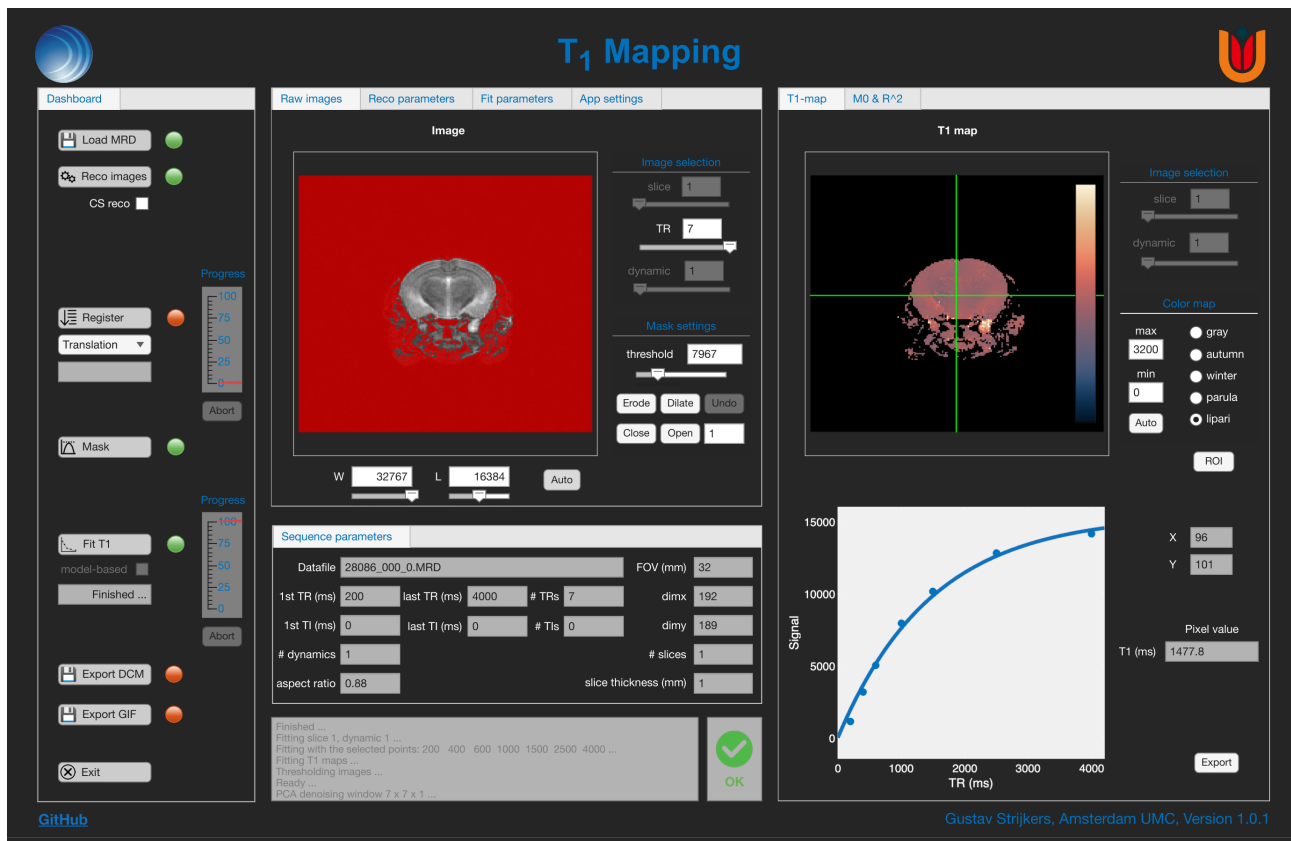


Step 8: Saturation recovery and variable flip angle experiments

The app also analyzes saturation recovery data, as shown in the figure below for a mouse brain. The data need to contain an array of TR values.

T1 is obtained by fitting signal intensity as function of TR to the following equation:

$$\text{Signal} = A(1 - \exp(-TR/T_1))$$



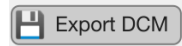
Variable flip angle series (spoiled gradient echo, the flip angles of the scan's VFA array) are fitted per pixel as a straight line through $S/\sin \alpha$ against $S/\tan \alpha$ (DESPOT1), whose slope is $\exp(-TR/T_1)$. For these series and for saturation recovery, the sliders and the selection lists are named after the TRs or the flip angles; flip angle series are fitted per cardiac frame, without B1 correction.



Step 9: Exporting the T1 maps

There are two ways to export the T1, M0 and R² maps.

(1) Export DCM



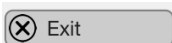
Exports the maps in Dicom and Nifti format for further processing in 3rd party software, and the T1 map also as a BIDS quantitative map. The program looks for the scanner's Dicom files of the scan and takes their header; if they are not found, the header, with the image position and orientation, is made from the MRD file.

(2) Export GIF

Exports the T1, M0 and R² maps as animated GIF files, with two pictures of the app's window.

When the MRD file lies in the scanner's folders, .../MRD/<study>/<scan>, the exports go to .../Data/<study>/DICOM, Nifti, BIDS and GIF, each into a new numbered folder; otherwise the app asks for a folder. The BIDS map is in seconds. The ROI data of Step 7 go to a folder of your choice.

Step 10: Exit

Press  to shut down the program.